Informatica® Quick Reference

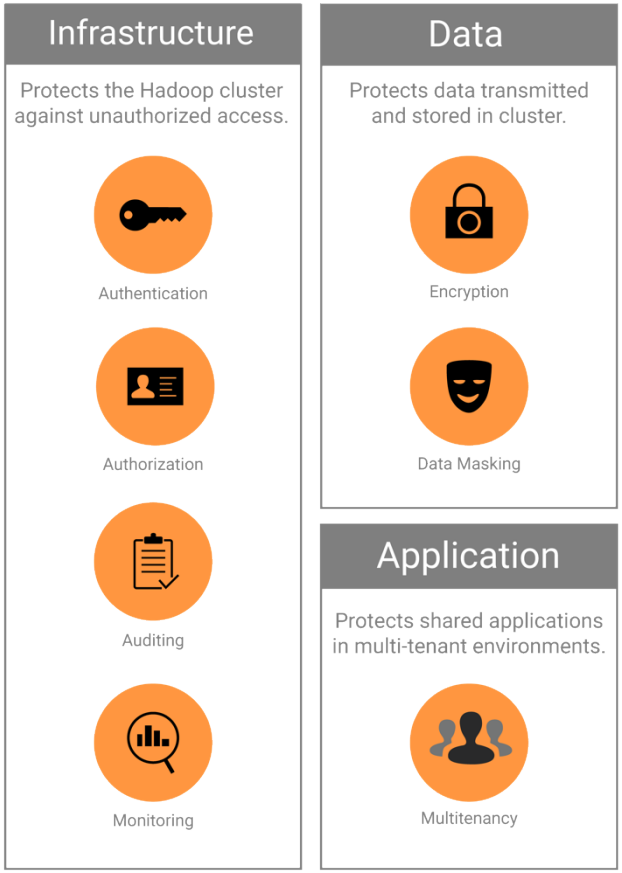
Big Data Management® 10.2.2 Hotfix 1: Security for Azure HDInsight

In this article, learn the security functionalities that Big Data Management supports and how to configure them in a Microsoft Azure HDInsight environment.

Overview

Informatica Big Data Management provides a holistic support for the security layers of the Hadoop cluster in order to keep your data and your enterprise secure. To integrate Big Data Management with a secure Hadoop cluster, it is not mandatory to configure security on the Informatica domain.

The following image shows different security aspects within each security layer:



This article discusses Big Data Management support for authentication and authorization mechanisms that an Azure HDInsight cluster uses.

Supported Security Systems for Azure HDInsight Clusters

The Spark and Blaze engines have identical supported security systems. The following table lists the supported security systems for the run-time engines in the Hadoop environment:

Authentication	Authorization
- Kerberos[1] - LDAP[2]	Apache Ranger
Enterprise Security Package[3]	

1. Use Active Directory-based Key Distribution Center for Kerberos.
2. Azure uses Azure Active Directory (AAD) to import users to a secure LDAP domain. See Azure Active Directory for Secure LDAP Authentication.
3. The Enterprise Security Package uses Kerberos and Ranger together for authentication and authorization.

Documentation Reference

For more information, refer to the following documentation:

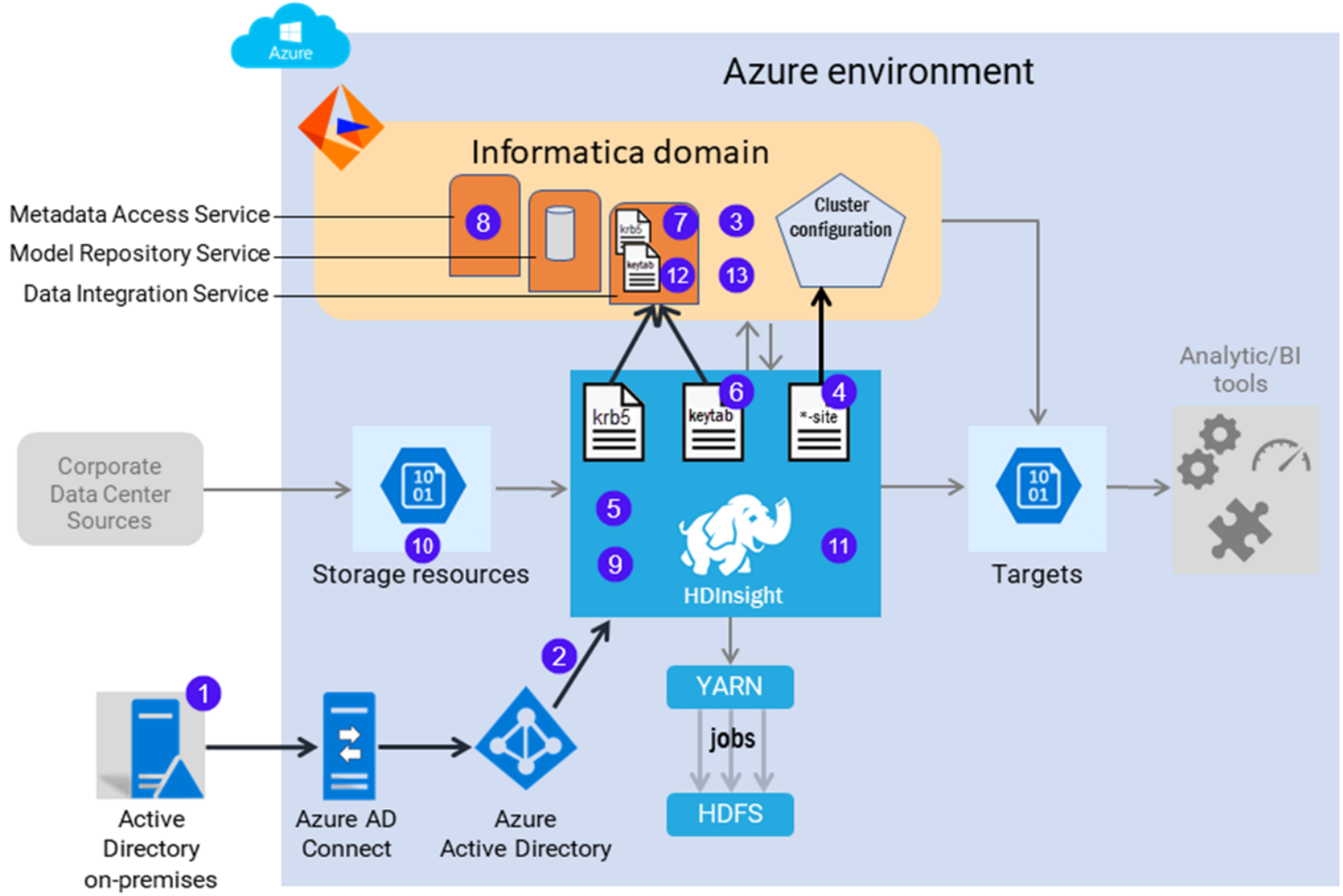
- HDInsight storage configuration documentation. Covers environment requirements to use ADLS with an HDInsight cluster.
- HDInsight security documentation. Covers HDInsight security concepts and configuration, including the Enterprise Security Package. Also see Azure Data Lake Storage (ADLS) documentation for Gen1 and Gen2[1] storage.
- Informatica 10.2.2 Hotfix 1 documentation:
  - The Big Data Management Integration Guide covers concepts and tasks to integrate Big Data Management with an HDInsight cluster.
  - The Big Data Management Administrator Guide covers Big Data Management security concepts and tasks.
  - The Informatica Security Guide covers Informatica domain security concepts and configuration, including information about integrating the domain with Azure Active Directory.

[1] Access to ADLS Gen2 storage is supported beginning with Big Data Management version 10.2.2 Hotfix 1 Service Pack 1.

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Security Architecture for Azure HDInsight

The following image shows the security architecture of a Big Data Management integration with an HDInsight cluster that uses Azure Active Connect and Azure Active Directory.



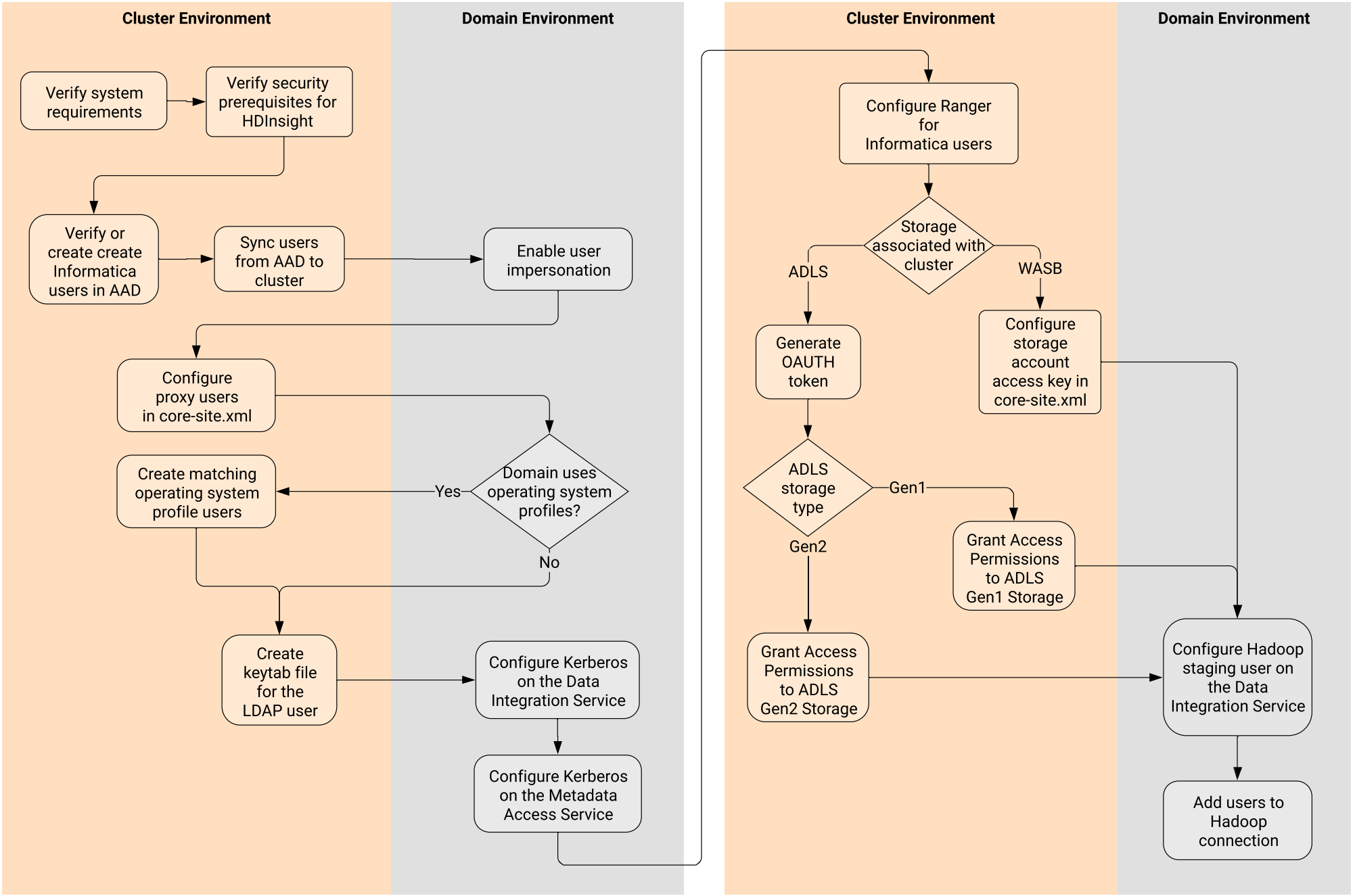
The following list of steps matches the numbered elements in the image and shows where to perform each step:
Note: Not all steps are mandatory, depending on your environment. See the flowchart on the next page for more information.

1. Verify that Informatica users exist in Azure Active Directory (AAD). (Users can be created in Azure AD or on-premises AD.)
2. Sync users from AAD.
3. Enable user impersonation on the domain.
4. Configure proxy users in the cluster core-site settings.
5. If applicable, create cluster users corresponding to OS profile users.
6. Create a keytab file for the LDAP user.
7. Configure Kerberos on the Data Integration Service.
8. Configure Kerberos on the Metadata Access Service.
9. Configure Ranger on the cluster.
10. Configure access to storage resources.
11. Generate the OAUTH token to access ADLS resources.
12. Configure Hadoop staging user on the Data Integration Service.
13. Add impersonation user and Blaze user to the Hadoop connection.

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Security Configuration for Azure HDInsight

The following flowchart shows the Big Data Management security configuration tasks and where to perform the tasks. Click on the task to see the corresponding topic in documentation.



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